

Turabi Yıldırım

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AI Researcher & Computer Engineer

Computer Engineering graduate focused on visual AI, and VLLM-based understanding. Experienced in building PyTorch models from scratch, experimenting with data augmentation, analyzing model results, contributing to research papers, and prototyping AI-agent systems. Currently pursuing an M.Sc. in Computer Engineering at Istanbul Technical University.

SKILLS

Languages: Python, C/C++, C#, Java, JavaScript, TypeScript, MySQL

AI/ML Tools: PyTorch, TensorFlow, Hugging Face, Weights & Biases, OpenCV, LangChain, LangGraph, LangSmith, MCP

Software & Tools: Docker, Git, Linux, ReactJS, Spring Boot, MongoDB, Hadoop

Strengths: Strong persistence and creativity

WORK EXPERIENCE

Research Student Jun 2023 - Feb 2025

Yıldız Technical University - TÜBİTAK-Supported Research Group

- Contributed to a TÜBİTAK-supported research project by developing PyTorch models from scratch, testing data-augmentation strategies, and supporting manuscript writing; helped the work get published in JSTARS (Q1, IF: 5.3).
- Trained LLaVA-OneVision on multiple remote-sensing datasets for a graduation project, producing one of the strongest models in its setting and presenting the project at the SIU 2025 conference.

Intern Jul 2025 - Aug 2025

ASELSAN

- Gained hands-on experience with AI-agent architectures, agent-based systems, and MCP.
- Built a rapid unit-test generation prototype using LangChain, React, and WebSocket.
- Strengthened software-development and teamwork experience in an enterprise environment.

AI Researcher Feb 2026 - Present

Odine-Supported Research Project

- Conducting research and development on AI-assisted video generation.
- Working on AI-based methods for activity detection.

EDUCATION

M.Sc. in Computer Engineering Feb 2026 - Present

Istanbul Technical University
Current GPA: 4.0/4

B.Sc. in Computer Engineering Sep 2021 - Aug 2025

Yıldız Technical University
GPA: 3.4/4

- Related coursework: C, Java, Object-Oriented Programming, Databases, Calculus, Computer Networks, Machine Learning, Data Structures and Algorithms, Assembly Programming, Microprocessor Systems.

Computer Engineering - ERASMUS Exchange Program Jan 2025 - May 2025

University of Oviedo

- Related coursework: C#, Java, Spring Boot, Object-Oriented Programming, Programming Paradigms, Compiler Design.

PROJECTS

LLaVA-RS (2025)

- Developed a satellite-image interpretation Vision-Language Large Model based on LLaVA-OneVision, integrating Qwen-based LLMs with visual encoders such as SigLIP.
- As part of my graduation project, achieved a literature-leading score on a remote-sensing change-captioning dataset under contemporary evaluation protocols.
- Presented the work at SIU 2025 and released one of the models used in the study as a Hugging Face [🔗](#) model repository.

MModalCC (2024)

- Trained vision-to-text models on multispectral and RGB satellite imagery; systematically evaluated model performance through different visual data augmentation strategies.
- Played an active role in developing the original model architecture and conducting experimental designs over a two-year research period; contributed to the analysis of hundreds of experiments across 10+ models.
- The study was published in IEEE JSTARS, a respected journal in the remote sensing field. *Q1, IF: 6.3*.

CERTIFICATES & ACHIEVEMENTS

- Adana Teknofest 2024 - Trendyol E-Commerce Competition Finalist
- Ankara Teknofest 2023 - Deneyap Robotics Competition Finalist
- BAYOYO TED University Artificial Intelligence Summer Camp
- ALES: 93.96 · TOEFL: 94 · YÖKDİL: 87.5
- Advanced MySQL Topics - Meta
- The Complete JavaScript Course 2022 - Udemy
- Machine Learning - Andrew Ng, Stanford / Udemy

REFERENCES

Available upon request.